

DANDY FIRMANSYAH

Senior Full Stack / Backend Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years of experience building, scaling, and modernizing high-throughput web platforms, REST APIs, and cloud services. Expert in TypeScript (Node.js), Golang, and relational databases (PostgreSQL/MySQL), with a proven track record of refactoring complex monoliths, optimizing serverless pipelines, and elevating unit test coverage from 10% to 75%. Experienced in guiding engineering teams, enforcing architectural standards, and delivering scalable end-to-end features.

CORE SKILLS

- **Full Stack & Core Stack:** TypeScript, JavaScript, Node.js, Golang, .NET Core, PHP
- **Backend & Cloud Services:** RESTful APIs, AWS (Lambda, RDS, S3), GCP (Cloud Run, Pub/Sub), Serverless, Microservices, Event-Driven Architecture
- **Databases & Caching:** MySQL, PostgreSQL, RDS, MongoDB, Redis, SQL Server
- **Frontend & Frameworks:** Vue.js, Angular Fundamentals, Ionic/Capacitor Awareness, Responsive Web Design
- **DevOps & Observability:** Docker, OpenTelemetry, CI/CD Pipelines, Automated Tracing, Git
- **Engineering Practices:** Unit Testing (Jest/GoTest), Clean Architecture, Distributed Systems, SNAP API Integration, Technical Mentorship

PROFESSIONAL EXPERIENCE

Batumbu Fintech | Senior Backend Engineer | Nov 2024 – Present

- **Domain Architecture & System Overhaul (SSOT):** Spearheaded a core system overhaul on the Loan Management System (LMS), removing redundant master data (Borrowers, Partners, Facilities, Virtual Accounts) to enforce a Single Source of Truth

(SSOT); refactored cross-domain SQL joins into optimized service-to-service calls without adding API latency, significantly reducing PostgreSQL memory usage and storage overhead.

- **Go Architecture & Monorepo Migration:** Co-architected and initiated the transition from a multi-repo structure to a unified Golang Monorepo alongside the Tech Lead; established shared internal packages (common utils, standardized HTTP/cURL clients, unified middleware) to eliminate code redundancy across microservices and streamline cross-team development velocity.
- **Architecture & Infrastructure Optimization:** Spearheaded the migration of asynchronous messaging pipelines from legacy RabbitMQ to GCP Pub/Sub, reducing RabbitMQ infrastructure dependencies by ~80% while enhancing tracing visibility and throughput during high-volume bulk loan extensions.
- **System Design & Scaling (Calculation Engine):** Redesigned the core interest and latefee calculation architecture from a bottlenecked, timeout-prone nightly batch scheduler to an on-demand, on-the-fly execution engine. Eliminating 100% of redundant batch processing and guaranteeing 100% calculation accuracy at scale.
- **Observability & Automation:** Researched, designed, and executed an autoinstrumentation tracing framework across Golang microservices, eliminating 100% of manual code spans and streamlining distributed tracing across all backend services.
- **Database Optimization & Tech Debt:** Led the deprecation of legacy transactional MongoDB schemas by consolidating data into PostgreSQL; eliminated unused schemas, unified infrastructure maintainability, and reduced database management overhead.
- **Engineering Quality & Standards:** Re-engineered low-quality vendor codebase for LMS; introduced rigorous unit testing standards and led the team to elevate code coverage from 10% to 75%, drastically reducing edge-case bugs and production regressions.
- **Internal Tooling & Operations:** Spearheaded the development of an internal operational suite, automating complex on-call workflows (loan backdating, settlement/repayment cancellations) to eliminate risky manual database queries and accelerate incident handling.
- **Engineering Operations & Governance:** Introduced and designed an automated Checker-Maker database query workflow integrated with Google Chat, eliminating manual access bottlenecks for on-call engineers, accelerating hotfix deployments, and establishing a full audit log for production data changes.

- **Banking & Core Systems:** Engineered core Loan Origination (LOS) and Loan Management (LMS) services using Golang and Node.js, integrating standardized SNAP banking APIs to streamline cross-institution financial operations.

Meratus Group | Technical Lead / Senior Fullstack Engineer | Feb 2024 – Oct 2024

- **0-to-1 System Architecture:** Led the technical design and initial development of the flagship enterprise SuperApp, establishing end-to-end backend architecture using .NET Core, Vue.js, MongoDB, and GCP to centralize logistics services and order booking.
- **Technical Leadership & Governance:** Acted as Tech Lead across a team of 5 engineers (Fullstack & Mobile); authored initial Technical Design Documents (TDD) and collaborated with BA teams on PRD specifications, introducing standardized documentation practices to the organization.
- **Cross-Functional ERP Integration:** Partnered with the internal ERP engineering team to design and build synchronized API contracts, mitigating integration risks and unblocking feature delivery between legacy ERP services and the new SuperApp portal.
- **Security & Infrastructure Setup:** Designed secure credential and environment variable workflows by integrating HashiCorp Vault for secrets management across GCP cloud deployment environments.

Fliptech Lentera Inspirasi Pertiwi | Software Engineer Backend | Oct 2021 – Jan 2024

- **Rapid Career Progression:** Earned promotion to Mid-Level Backend Engineer within 6 months based on exceptional feature delivery, technical ownership, and rapid adoption of engineering standards.
- **Monolith Decomposition & Microservices:** Partnered with Staff Engineers to initiate and build a dedicated Golang microservice from scratch for the Digital Products domain, successfully decoupling high-volume catalog/product management out of the core legacy PHP (Yii2) monolith.
- **API Performance & Caching Optimization:** Re-architected the international transfer rate-calculation engine; replaced heavy, on-the-fly multi-country margin loops with an optimized Redis caching strategy, slashing API response latency from 4–8 seconds to under 1 second (<800ms) as country support scaled.
- **High-Availability Financial Features:** Developed and maintained mission-critical third-party payment integrations, ensuring high reliability across highconcurrency international payment and transfer pipelines.

- **System Reliability & Operational Support:** Investigated and resolved critical production incidents, optimized query performance across MySQL and Redis layers, and actively reduced technical debt to maintain system stability.

PT Multi Guna Gemilang | Full Stack Developer | Oct 2020 – Oct 2021

- **Real-Time Logistics Applications:** Designed and engineered messaging and shipment tracking applications for enterprise logistics clients, enabling real-time communication via WebSocket technology.
- **Full-Stack Development:** Built scalable backend services using Node.js and MongoDB alongside reactive frontend dashboards using Vue.
- **DevOps & Delivery:** Managed containerized application deployments using Docker and automated CI/CD pipelines in GitLab to ensure seamless release cycles.

PT Electronic Data Interchange Indonesia | Programmer | Jun 2016 – Jan 2020

- **Enterprise & Customs Applications:** Developed and maintained core business applications and government customs-related software across multiple enterprise clients.
- **Operational Process Automation:** Automated complex manual business processes, significantly improving operational efficiency and data accuracy.
- **Reporting & Multi-Database Systems:** Built executive reporting dashboards and managed database schemas across Oracle, PostgreSQL, SQL Server, .NET, Java, and PHP environments.

EDUCATION

Bachelor of Information Systems
Universitas Mercu Buana